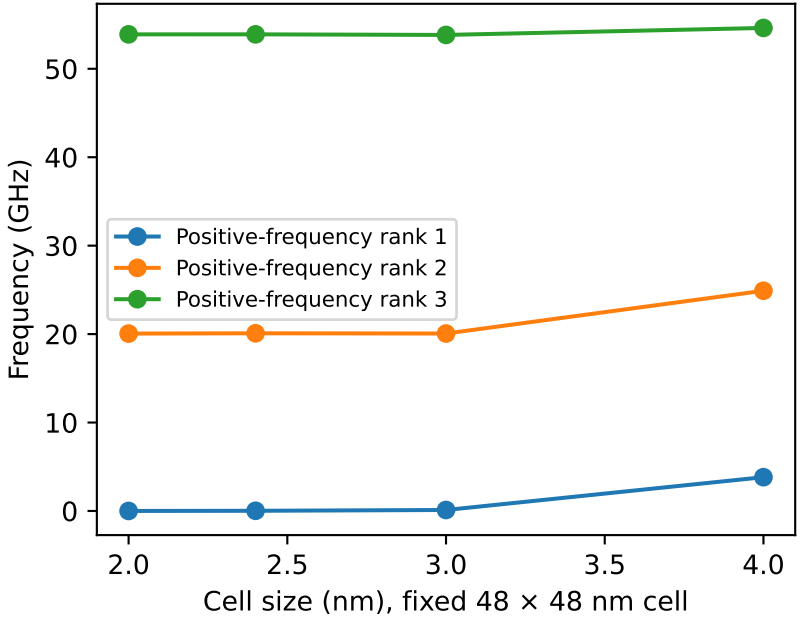
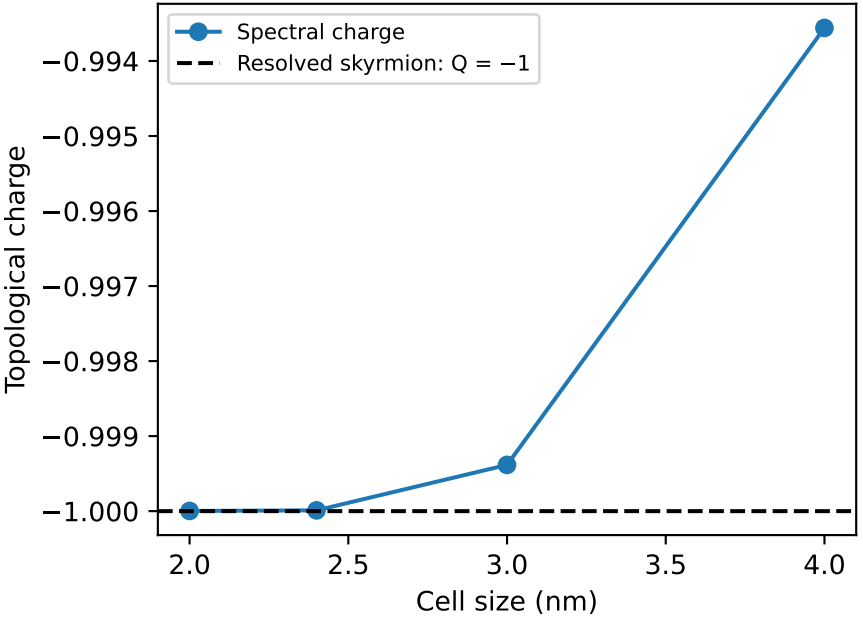

Frequency (GHz)
50
40
30
20
10
0
Positive-frequency rank 1
Positive-frequency rank 2
Positive-frequency rank 3
2.0
2.5
3.0
3.5
4.0
Cell size (nm), fixed 48 × 48 nm cell


Topological charge
−0.994
−0.995
−0.996
−0.997
−0.998
−0.999
−1.000
Spectral charge
Resolved skyrmion: Q = −1
2.0
2.5
3.0
3.5
4.0
Cell size (nm)